EHR Media Intelligence Platform: technical write-up

Local assessment demo. Synthetic data only. Commit history on `main`: one commit per major task (foundation + ingestion, FHIR R4, AI summaries, semantic search, clinician UI) plus a quota-error fix.

Read first. (1) **Live OpenAI verification is a single successful request** (synthetic fixture, one attempt, HTTP 200, 975 tokens): it shows the integration works once, not model reliability. (2) **Clinical correctness has not been independently validated**: no clinician has reviewed any output, and grounding checks show textual traceability only. (3) **Authentication and authorization are not implemented.** (4) **Synthetic-data demo only**: do not use with real patient data; not HIPAA-assessed.

What it does

Ingests JSON/CSV/text records; normalizes dates, gender and MRN; detects duplicates and identity conflicts; maps each resolved patient to FHIR R4 (Patient, DocumentReference, DiagnosticReport, Bundle); produces grounded AI summaries (OpenAI behind an interface); supports semantic search; and serves a React clinician search UI.

Architecture

Clean Architecture in a modular monolith (FastAPI): `domain` (rules, ports) ← `application` (use cases) ← `infrastructure` (SQLAlchemy/SQLite, OpenAI, sentence-transformers, fhir.resources) and `api` (thin routes). An import-direction test enforces the layering, and that `fhir.resources` and `openai` stay inside `infrastructure`. SQLite is the source of truth; the vector index is derived and rebuildable. Patterns: Repository/Unit of Work, Strategy (3 ingestion formats, 2 embedders), one Factory, Adapters, DI.

Key decisions (all in `docs/DECISIONS.md`, D1–D43)

- **Conservative identity:** never merge on name similarity; same MRN + different DOB = conflict; same name+DOB under a new MRN = ambiguous → held for review, with rationale and audit entry.
- **FHIR R4 concepts:** per patient a *collection* Bundle of Patient, DocumentReference (note text as a base64 attachment; `context.period` = clinical date, `date` = ingestion time) and DiagnosticReport, linked by `urn:uuid` references with deterministic ids. Uses `fhir.resources.R4B` (FHIR 4.3.0, a technical correction of R4 whose elements for these resources were checked as identical to 4.0.1); the package default namespace is R5 and is never used. The library does not enforce value sets or reference resolution, so validation has three layers (library, value-set checks, reference/attachment checks). FHIR state is separate from ingestion status; generation is idempotent by fingerprint. Not a complete FHIR validator.
- **Grounded summaries:** the model sees only one patient's clinical text (no name, MRN, DOB, id); every statement needs a verbatim evidence quote and source record ids; deterministic checks cover cited ids, quotes, diagnosis-in-quote, hedging/history/negation, dates/numbers, advice, ≤200 words. Cache key: patient + content hash + prompt version + model. A quote proves traceability, not correctness.
- **AI summary evaluation:** 18 deterministic tests on synthetic cases (unsupported claims, dates, identity, attribution, length, schema) plus a human-review rubric that has **not** been performed; English-only heuristics; the quality indicator is a coverage heuristic. Live result: one request passed all automated checks; no clinician review.
- **Search:** SQLite vector store (exact cosine, filters applied before ranking, every hit re-validated) instead of ChromaDB; default `all-MiniLM-L6-v2`; the lexical embedder is an explicit fallback and there is never a silent substitute (unavailable semantic → `503`). Results carry no names/MRNs; the UI fetches them from a separate `GET /patients/{id}/display`, and detail views use a focused, ownership-checked FHIR resource endpoint rather than the whole Bundle.
- **Trade-offs:** SQLite exact-scan vectors over ChromaDB (no extra service, rebuildable, but linear: 130 ms at 20,000 entries, unmeasured beyond); semantic default with an honest lexical fallback (a 1 GB install and 503 until set up, in exchange for never silently degrading); explicit per-patient FHIR/summary/index steps (simple, idempotent, retryable, but not automatic); no authentication, to keep the demo scope honest rather than add a cosmetic mechanism.
- **Error handling:** provider/SDK text is never propagated or logged; HTTP 429 `insufficient_quota` is permanent (one attempt), rate limits/5xx/network errors use bounded retry.

Verification actually performed

Check	Result
Backend `pytest` / `ruff`	503 passed, 1 opt-in skipped (real-model test); lint clean. Also passes from a clean `git archive` copy on Python 3.14 (base `[dev]` install); run in the working tree on Python 3.12.
Frontend	24 tests (mocked API) passed; production build OK; `npm ci` clean copy OK.
Live OpenAI (one request)	Real API, `gpt-4o-mini-2024-07-18`, HTTP 200, exactly 1 provider attempt, 975 total tokens. Passed schema, 46 words (≤200), grounding (cited ids exist, quotes verbatim in cited records), disclaimer, persistence. Second POST = cache hit, no second provider call. A live summary was then indexed and shown in the React UI (badge, banner, quotes, disclaimer). An earlier attempt had failed (likely quota) before the quota fix.
Real-browser smoke (Chrome, macOS, one run)	23/23 checks: ingest → FHIR → summary (*local stub, not OpenAI; the live call was a separate test*) → index → search, cards, name/MRN lookup, both drawers, filters, no overflow at 768/375 px.
Search benchmark (synthetic, 1 run)	Semantic: cold start 3.7 s (model on disk; ~50 s first ever), index 50 records 0.38 s, warm search p50/p95 8.0/8.8 ms; lexical 3.1/4.0 ms. 20 author-written queries: top-1 1.00 / precision@5 0.89 (semantic) vs 0.75 / 0.45 (lexical).
Hygiene	Mutation checks on safety properties; secret/private-file scan of tracked files and full history: only fake fixtures.

Limitations and deviations

- **Not verified:** model reliability or output quality beyond one live request; clinician review of any summary; Windows setup commands; other browsers; automated accessibility/visual tests; contrast.
- **Security:** no authn/authz; any caller can read all data; the full-Bundle FHIR endpoint returns whole Patient resources; patient filtering is data scoping, not access control.
- **Deviations from the brief:** SQLite vector store instead of ChromaDB; no `GET /patients` or `GET /patients/{id}` (display lookup instead); search field is `limit` (max 5), not `top_k`; FHIR/summary/index are explicit per-patient calls, not automatic after ingestion.
- **Other:** English-only heuristics; quality indicator is a coverage heuristic, not a confidence; no `needs_review` workflow; no migrations; no CI.